DESCRIPTION OF COURSEWORK

Course Code	EEE 323
Course Name	Digital Signal Processing
Lecturer	Ts. Dr. Kang Chia Chao
Academic Session	04-2024
Assessment Title	Assignment

A. Introduction/ Situation/ Background Information

Digital signal processing, more commonly known as DSP, is a field of study with increasingly widespread applications in our technological world. The incredible advances in digital signal processing technologies over the last decade along with the advent of fast and economical analog-to-digital converters (ADCs) have developed a strong interest in cognitive radio for both commercial and military applications. A discrete-time system is an entity that processes a discrete-time input signal to produce a discrete-time output signal. The digital signals processed in this manner are a sequence of numbers that represent samples of a continuous variable in a domain such as time, space, or frequency. DSP applications include audio and speech processing, sonar, radar and other sensor array processing, spectral density estimation, statistical signal processing, digital image processing, data compression, video coding, audio coding, image compression, signal processing for telecommunications, control systems, biomedical engineering, and seismology, among others. DSP can involve linear or nonlinear operations. Nonlinear signal processing is closely related to nonlinear system identification and can be implemented in the time, and frequency domains. The application of digital computation to signal processing allows for many advantages over analog processing in many applications, such as error detection and correction in transmission as well as data compression. Digital signal processing is also fundamental to digital technology, such as digital telecommunication and wireless communications. On the other hand, MATLAB and signal processing products help us to analyze signals from a range of data sources. We can acquire, measure, transform, filter, and visualize signals without being an expert in signal processing theory.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 2 Analyze complex engineering problems in digital signals and systems by applying discrete

signal processing approach.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalised by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

- Group Assignment
- Copy-paste is prohibited and will significantly degrade your marks.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
	CLO 2	
1.	Introduction	20
2.	Matlab Content	15
3.	Appearance and formatting	10
4.	References	5
5.	Overall Content	15
6.	Organization: Overall	15
7.	Conclusion	20
	TOTAL	**100**

F. Task(s)

SAMPLING THEOREM

The sampling theorem specifies the minimum-sampling rate at which a continuous-time signal needs to be uniformly sampled so that the original signal can be completely recovered or reconstructed by these samples alone. This is usually referred to as Shannon's sampling theorem in the literature. Your task is to make a report on:

Objective	The objective of this study is to demonstrate and verifying sampling theorem. This assignment is to be done in Matlab.
Problem Statement	The manufacturing sector has instituted some major changes in factory floor operations over the last several years. These changes include the use of robotics, just-in-time inventory management, computer aided design, and computer aided manufacturing. As an engineer attached to the manufacturing company, you are asked by the senior engineer to construct and demonstrate the data collection process for cyclic operations in a factory environment from the point of view of information theory. This conceptual approach uses the Nyquist Sampling Theorem criteria to identify the impact on accuracy of the data collection process. **PART 1** The design criteria is generate a sinusoidal wave of 1kHz, determine the Nyquist frequency and verify the sampling theorem, **PART 2** Showing output waveforms for undersampled, oversampled and right sampled cases.
Ways and Means	**Tools** The group must identify the availability/suitable of the selected simulation tools. **Procedure** The group is required to search for the relevant step/information to carry out the study based on the appropriate guideline or manual provided by the lab. **Data Acquisition** All data collected are observed and analysis during the study must be tabulated in the proper format for easy verification and presentation of the technical report.

Results	The group is required to submit a technical report within one month. In the report, you should include the introduction, the proper procedure, analysis of data and the outcome of the study.

NOTE

1) YOUR REPORT (PDF) MUST HAVE COVER PAGE, INTRODUCTION, CODING & RESULTS, DISCUSSION CONCLUSION, REFERENCES.

2) MIN PAGES 10 PAGES. KINDLY COPY THIS RUBRICS IN YOUR REPORT.

3) PER GROUP ONLY ONE PERSON UPLOAD TO MOODLE

DUTE DATE: 28 MAY 2024, 12.00PM

RUBRICS

Component Title	Assignment Report				Percentage (%)	100%	
	Score and Descriptors					Weight (%)	Marks
Criteria	Very Good 5	Good 4	Average 3	Poor 2	Very Poor 1		
	CLO 2: Analyze complex engineering problems in digital signals and systems by applying discrete signal processing approach						
Introduction	Introduction complete and well-written; provides all necessary background principles for the experiment	Introduction nearly complete, clear missing some minor points	Introduction is nearly complete, missing some minor points	Some introductory information, but still missing some major points	Very little background information provided or information is incorrect	**20**	
Matlab Content	Exceptionally well-presented and argued; ideas are detailed, well-developed, supported with specific evidence & facts, as well as examples and specific details about DSP and continuous time signal and system.	Well-presented, argued; ideas are detailed, developed and supported with evidence mostly specific about DSP and continuous time signal and system.	Well-presented ideas, developed and supported with some evidence of fundamental knowledge about DSP and continuous time signal and system.	Content is sound and solid; ideas are present but not particularly developed or supported; some evidence about digital signal processing, but usually of a generalized nature.	Content is not good and no fundamental knowledge about DSP	**15**	
Appearance and formatting	All sections in order, well-formatted, very readable	All sections in order, formatting generally good but could still be improved	Some sections in order, formatting generally good but could still be improved	Sections in order, contains the minimum allowable amount of handwritten copy, formatting is rough but readable	Sections out of order, too much handwritten copy, sloppy formatting	**10**	
References	Sufficient, consistent and logical referencing format and citation	Acceptable number of referencing and consistent referencing format.	Inconsistent referencing format and citation with minor mistakes	Inadequate referencing and citation.	References are not included	**5**	

Overall Content	Exceptionally well-presented and argued about analyze complex engineering problems in digital signals and systems by applying discrete signal processing approach	Well-presented, argued when analyze complex engineering problems in DSP by applying discrete signal processing approach.	Content CLO 2 is sound and solid but still can be improved in analyze complex engineering problems in DSP by applying discrete signal processing approach.	Content CLO 2 is sound and solid; but not particularly developed or supported when analyze complex engineering problems in DSP by applying discrete signal processing approach.	Not well - presented and argued about analyze complex engineering problems in DSP by applying discrete signal processing approach	**15**	
Organization: Overall	Well-planned and well-thought out. Includes title, introduction, statement of main idea, transitions and conclusion**.**	Good overall organization, includes the main organizational tools.	Average overall organization, includes the main organizational tools.	There is a sense of organization, although some of the organizational tools are used weakly or missing	No sense of organization	**15**	
Conclusions	All important conclusions have been clearly made, student shows good understanding	All important conclusion have been drawn, could be better stated	Some important conclusion have been drawn, could be better stated	Conclusions regarding major points are drawn, but many are miss stated, indicating a lack of understanding	Conclusions missing or missing the important points	**20**	
					TOTAL		